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An Engineering Methodology for Solar Thermal System Design in Buildings Aligned with the ISO 50001 Planning Framework

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Abstract

This study presents an integrated engineering methodology aligned with the planning phase of the ISO 50001:2018 (Energy Management Systems—Requirements with Guidance for Use. International Organization for Standardization (ISO): Geneva, Switzerland, 2018) energy management standard for the design, sizing, and assessment of a solar thermal system applied to domestic hot water production in a medium-scale hotel building. The proposed framework focuses on the energy review stage of ISO 50001, incorporating site-specific climatic assessment, spatial layout optimization, structural feasibility analysis, and energy performance evaluation to support informed technology selection and system viability. Thermal performance is assessed using real operational data from the case study, complemented by a data-driven multivariable regression-based energy performance indicator (EnPI) that relates electricity consumption to cooling degree days and room occupancy. This regression model, developed in accordance with ISO 50001 recommendations, enables transparent monitoring of energy performance under real operating conditions without relying on black-box predictive techniques. Material selection criteria for absorber plates, heat-transfer components, transparent covers, and insulation layers are discussed to support both initial efficiency and performance stability under site-specific climatic conditions. In addition, an indicative and qualitative analysis of material-dependent performance evolution is introduced to support comparative decision-making, without implying quantitative lifetime prediction. Structural feasibility of the collector support system is examined through finite-element simulations under combined gravitational and wind loads, providing illustrative verification of stress distribution under representative operating conditions. The installed system delivers an annual thermal energy contribution of 8468 kWh, resulting in an estimated reduction of 7.79 t of CO_2 emissions per year. Economic indicators suggest a short payback period and a favorable internal rate of return, which should be interpreted as order-of-magnitude estimates within the planning scope of the methodology. Overall, the proposed methodology provides a replicable and multidisciplinary planning-phase framework aligned with ISO 50001 for the design and assessment of solar thermal systems in medium-scale buildings under real operating conditions.

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1. Introduction

Rapid scientific, technological, and economic growth has generated a growing dependence of humanity on fossil-fuel-based energies [1]. According to the International Energy Agency (IEA), worldwide fossil-fuel-based energy demand grew by 2.3% in 2019 [2]. Solar energy, promoted by international environmental treaties to curb pollution, has firmly established itself among other renewable energy technologies, comprising nearly 31% of the total installed renewable energy capacity in 2022, making it the second most installed renewable energy resource [3,4]. With the gradual depletion of fossil fuels and increasing environmental degradation, the development of new clean energy has become a key strategy in the global energy sector [5,6]. This growing demand poses a major challenge for the environment, as these conventional energies are responsible for pollution, high carbon footprints, and global warming [1,7,8]. Renewable energy systems make efficient use of energy resources that are replenished naturally and thus improve the sustainability of systems [9]. Solar energy capture technology has evolved over the years, and current approaches increase energy collection and achieve higher thermal efficiency [10]. In addition, solar power generation is expected to increase to 48% by 2050, driven by economic and industrial growth [11]. Solar thermal systems (STSs) harness solar energy to generate heat for domestic hot water and, in some cases, electricity [12]. In STSs, there are three types of systems that differ by application, fluid end use, and installation design: low-, medium-, and high-temperature [13,14]. Proper sizing and configuration for the specific application components of a solar water heating system is essential to reduce capital investment and ensure reliable system operation [9,14]. A low-temperature water-heating system is intended to heat water below the point of boiling at temperatures between 50 °C and 65 °C. A low-temperature STS is currently used in many countries as a clean, reliable, and cost-effective method for harnessing solar power to meet 50–80% of hot water needs in both commercial and residential sectors [2]. However, to design renewable energy systems that provide space heating and domestic hot water (DHW), it is paramount to quantify the building's heat energy consumption [15]. The performance of different technologies should be analyzed, taking into account other aspects such as local solar irradiation, heat load, and process temperature requirements, and working fluid inlet and outlet temperatures, surface requirements, number of hours the system can operate in a day, and initial investment costs [16]. Experimental and modeling studies have been conducted to determine the energy, economic, and environmental efficiency of thermal solar collectors operating under various climatic conditions [17]. For example, the study by Panagiotidou et al. [18] compared different configurations of solar water heating systems adapted to medium-height buildings in several cities in Greece.

Selecting a suitable location for the installation of solar power plants is essential to optimizing the efficiency of the system. Average annual solar irradiation is an environmental indicator used by countries for the selection of sites [19,20]. The Sun releases an enormous amount of irradiation energy: 174 PW (1 PW = 1015 W) in the upper atmosphere of the Earth [21], but due to its surface characteristics, the amount of irradiation is not constant. In countries located in the 'Sunbelt', there is enormous potential for solar energy, where there is an abundance of global horizontal solar irradiation, with an average of 1700 kWh/m^2 per year [11,22]. Control of system operation based on the occupant's hot water usage on the demand side affects the energy performance of STSs [23]. For example, hospitals

consume large amounts of energy to produce hot water and compensate for thermal losses from distribution and recirculation [24]. Properly assessing the demand for hot water is crucial for correctly selecting and sizing a solar thermal system, thus ensuring optimal techno-economic performance [25].

Several methodologies have been proposed for optimizing solar thermal systems. For instance, Sánchez [26] analyzed the effects of latitude and vacuum-tube orientation on system performance, emphasizing their impact on the achievable temperature range. O'Hegarty et al. [15], building on Lutz et al. [27], developed a model that enables rapid and graphical sizing of solar thermal systems. A key strength of this approach is using occupancy as a more accurate input variable than floor area for estimating domestic hot water demand. However, a common limitation of these methodologies is their narrow scope: they tend to focus either on thermal performance or on demand estimation in isolation, often overlooking structural constraints, roof geometry, or economic viability. Moreover, they generally lack validation through real-case implementations, which limits their applicability in complex, built environments. In contrast, the methodology presented in this study aims to address these gaps by integrating climatic, structural, spatial, economic, and environmental factors into a single decision-making framework. Its practical relevance is demonstrated through a case study in a medium-sized hotel in Cienfuegos, Cuba, where the proposed approach is applied, evaluated, and validated under real operating conditions.

As shown, several methodologies currently exist for optimizing the selection and sizing of solar thermal systems. However, these methodologies address only some of the factors to consider in proper sizing. This paper proposes a methodology for selecting and sizing solar thermal systems, accounting for the climatic and structural characteristics of residential areas. To determine the level of efficiency obtained—including economic and environmental sustainability criteria—variables such as geolocation, the area available for the installation, domestic hot water demand, the irradiation level of the area, the useful hours of collection, and the initial investment are considered to adequately size the system.

The main contributions of this study are summarized as follows:

- Integrated methodology for solar thermal system design: This study proposes a unified framework for selecting and sizing solar thermal systems, combining climatic, structural, energy, economic, and environmental criteria, thus overcoming the fragmented nature of previous approaches.
- Advanced simulation-based analysis: The methodology incorporates solar geometry modeling and structural simulations to accurately evaluate irradiance levels, shading effects, roof geometry, and mechanical stability. This ensures its applicability to diverse climates and building typologies worldwide.
- Comprehensive economic and environmental evaluation: The approach includes detailed financial and environmental assessments, enabling comparisons between technological alternatives, calculating investment returns, and estimating avoided CO_2 emissions, providing a globally adaptable decision-making tool for sustainable implementation.

To address current methodological gaps in the integration of solar thermal systems for medium-scale buildings, this study leverages simulation tools such as SketchUp Pro and SolidWorks (2021). These platforms enable precise assessment of solar exposure, shading patterns, and structural stability, which are critical for system viability. Additionally, the methodology incorporates multi-criteria decision analysis (MCDA) to evaluate technological options under diverse technical, economic, and spatial constraints. This integrative approach ensures a replicable and context-sensitive framework for solar system design.

2. Materials and Methods

The search focused on studies addressing solar thermal systems, sizing methodologies, and the integration of climatic, structural, economic, and environmental criteria for renewable energy applications. The selection criteria included peer-reviewed articles, publications from relevant conferences, and technical studies applied in real-world contexts, particularly in medium-scale buildings and hotel facilities.

2.1. Literature Review

To develop the methodological framework of this study, an exhaustive analysis of the scientific literature was carried out using a systematic search in recognized academic databases. The focus of the analysis was structured under the following search strategy:

```
TITLE-ABS-KEY("solar thermal systems" OR "solar water heating"
OR "thermal collectors")
AND TITLE-ABS-KEY("system sizing" OR "optimization"
OR "design methodology" OR "dimensioning")
AND TITLE-ABS-KEY("climatic conditions" OR "structural analysis"
OR "economic analysis" OR "environmental impact")
AND PUBYEAR > 2015 AND PUBYEAR < 2025
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This temporal filter limited the results to research published between 2016 and 2024, ensuring inclusion of the most recent and relevant advances in the field.

2.2. Methodological Context and Literature Integration

The analysis identified that the most commonly used approaches for optimizing solar thermal systems include thermal modeling, computational simulations, and multi-criteria decision-making methods. These strategies have been applied in the literature to improve collector efficiency, evaluate structural feasibility, and integrate cost-benefit assessments.

In this context, variables such as solar irradiance, domestic hot water demand, roof geometry, and structural conditions are regularly highlighted. Additionally, economic indicators such as investment, payback period, and life-cycle analysis are increasingly incorporated to assess project viability. These methodological insights were adapted for the present research, combining proven procedures with advanced computational tools such as SketchUp and SolidWorks to validate spatial and structural feasibility.

The novelty of this study lies in integrating technical, climatic, structural, economic, and environmental dimensions into a single framework—an approach that remains underexplored in the literature. Figure 1 presents the bibliometric network of perspectives and trends in solar thermal research. The trends indicate a shift from isolated performance studies toward multidisciplinary approaches that integrate optimization techniques, techno-economic analysis, and sustainability criteria. These findings align with the methodological approach of this research, reinforcing its relevance and originality.

2.3. Analysis of Previous Research on Solar Thermal Systems

The comparative analysis presented in Table 1 shows that although several studies on solar thermal systems (SSTs) in buildings have addressed aspects such as energy consumption, collector efficiency, or experimental performance improvements, most exhibit significant limitations. In particular, many studies overlook the influence of local climatic conditions, structural feasibility, or integrated economic and environmental assessments, often limiting their scope to partial analyses with limited applicability in real-world contexts. These gaps reinforce the novelty of the present research, which integrates climatic, structural, economic, and environmental criteria into a unified methodology. By validating the approach through simulations in SketchUp and SolidWorks and quantifying both

financial indicators and avoided CO_2 emissions, the proposed framework addresses the methodological shortcomings identified in previous studies, and provides a replicable model for implementing solar thermal systems in medium-scale buildings.

Table 1. Summary of studies on energy analysis, efficiency, and economic evaluation of solar thermal systems (SSTs) in buildings.

Ref.	Energy Analysis	Shading and Efficiency Analysis	Economic and Environmental Evaluation
[28]	Analyzes monthly electricity consumption without considering energy performance indicators.	Does not include irradiation magnitude or shading studies.	Evaluates only electricity savings and CO_2 emission reductions.
[29]	Analyzes daily electricity consumption without considering performance indicators.	Does not consider shading or inverter efficiency ranges.	Evaluates payback period, investment and maintenance costs, and reduction of CO_2 emissions.
[30]	Analyzes energy consumption during peak hours without performance indicators.	Does not consider shading or an efficient arrangement structure.	No economic or environmental evaluation.
[31]	None.	None.	None.
[32]	None.	Evaluates efficiency through experimental arrangements.	None.
[33]	Analyzes electrical and thermal behavior based on hot water profiles and ambient conditions.	Develops and validates a dynamic TRNSYS model for SSTs.	None.
[34]	None.	Evaluates two SST configurations with spherical turbulators, achieving a 15.77% increase in outlet temperature.	Payback period ranges between 2.98 and 3.51 years.
[35]	None.	Studies the effect of nanofluid mass flow rate on system efficiency.	Performs an integrated energy-exergy-economic analysis.

2.4. Integration of the ISO 50001 Energy Management Framework

ISO 50001 is structured around the continuous improvement cycle of plan–do–check–act (PDCA), as depicted in Figure 2, and integrates energy management into the daily operations of an organization. Within the framework of energy management, the PDCA cycle can be outlined as follows:

- Plan: Understand the organizational context, define an energy policy, and establish an energy management team. Assess actions to address risks and opportunities, conduct an energy review, identify significant energy uses (SEUs), and set up energy performance indicators (EnPIs), energy baselines (EnBs), objectives, energy targets, and action plans aimed at improving energy performance in line with the organization's energy policy.
- Do: Implement the action plans, operational and maintenance controls, and communication strategies, ensure the necessary competence, and consider energy performance in the design and procurement stages.
- Check: Monitor, measure, analyze, evaluate, audit, and conduct management reviews of energy performance and the energy management system (EnMS).
- Act: Take corrective actions to address nonconformities and drive continuous improvement in energy performance and the EnMS.

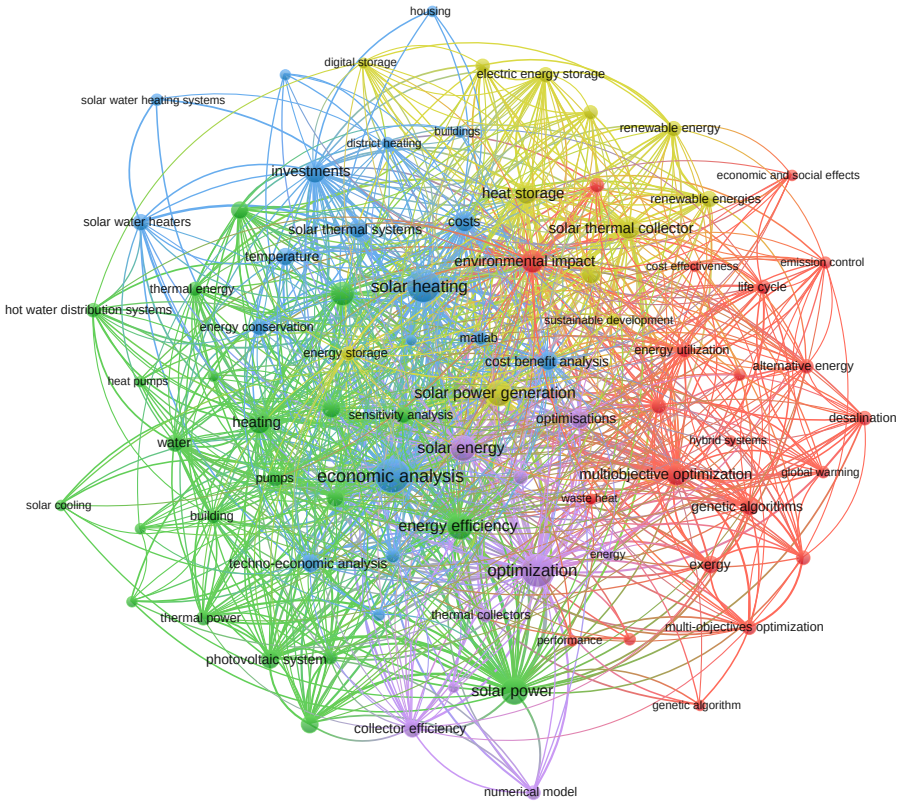


Figure 1. Bibliometric network of perspectives and trends in solar thermal systems and their optimization, highlighting the convergence of technological, economic, and environmental approaches.

As noted earlier, the primary focus of this research is on the planning phase, which is clearly reflected in Figure 2 as part of the ISO 50001 standard.

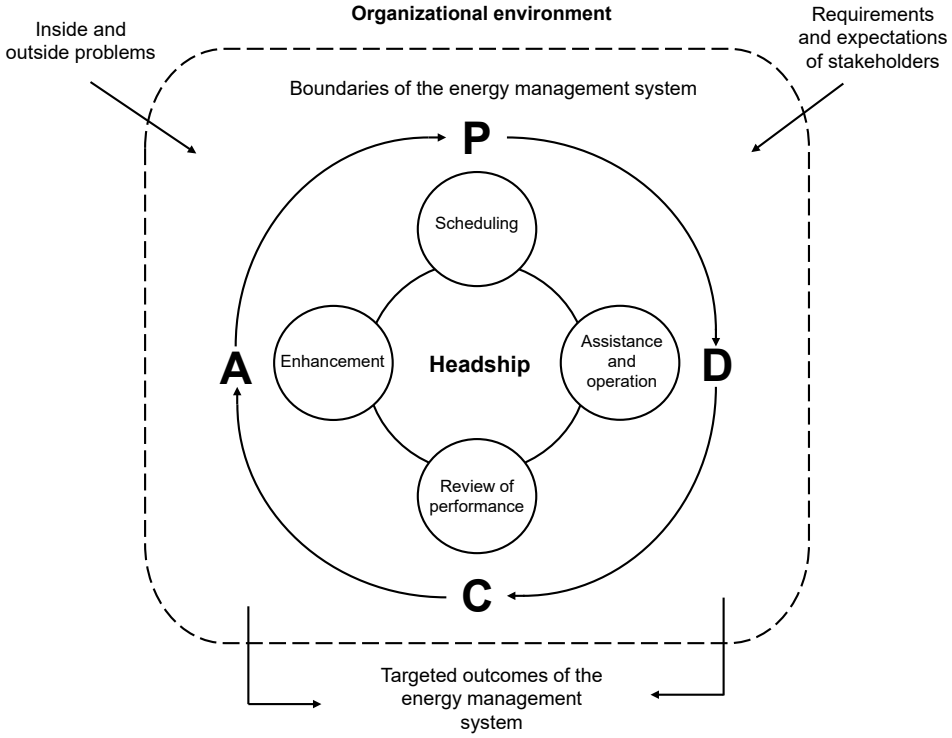


Figure 2. The PDCA cycle model (Plan, Do, Check, Act).

Also, it is important to note that this study does not aim to implement the complete PDCA cycle of ISO 50001. Instead, it deliberately focuses on the planning stage, which represents the most critical phase for technology selection, system sizing, and feasibility assessment in energy-intensive building services such as domestic hot water production. By concentrating on the energy review process, including the identification of significant energy uses, the definition of EnPIs, and the establishment of EnBs, the proposed methodology aligns with ISO 50001 while maintaining a clearly delimited and technically justified scope.

The approach adopted in this study is based on the planning phase of the ISO 50001 standard, as illustrated in Figure 3. This phase specifically emphasizes the energy review process, which is conducted to assess energy uses and consumption, pinpoint significant energy usage areas, and identify primary opportunities for energy improvements. The figure provides a synthesis of the standard at this particular stage, offering a clearer understanding of the methodology employed in this work.

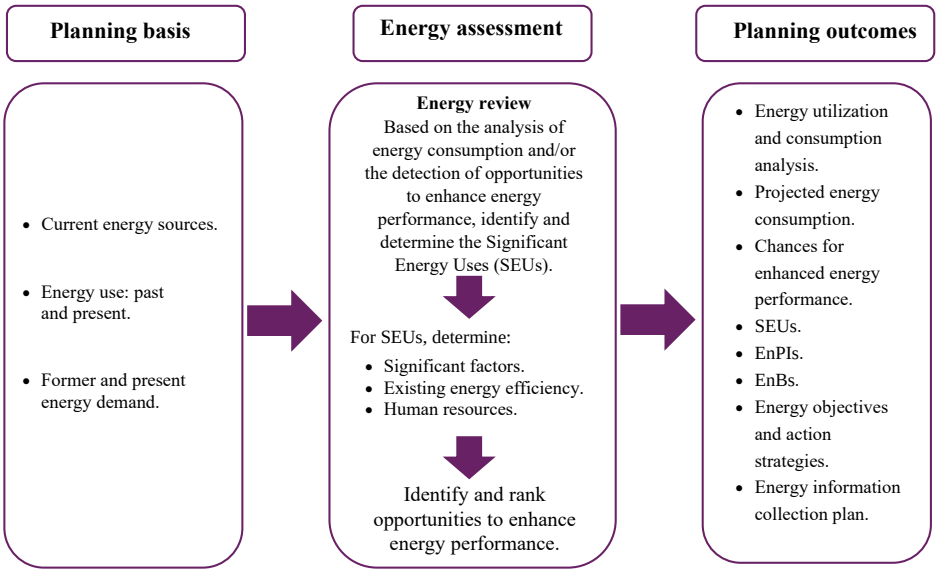


Figure 3. Methodology employed, based on the ISO 50001 standard.

Explicit Mapping of ISO 50001:2018 Planning Requirements to the Proposed Methodology

The proposed engineering methodology is explicitly aligned with the planning phase (Plan) of the ISO 50001:2018 standard, which follows the plan–do–check–act (PDCA) cycle. While methodological references to ISO 50001:2011 are retained for continuity with established practices for developing EnPIs and EnBs, the implementation and interpretation adopted in this study strictly adhere to the requirements of ISO 50001:2018.

Within this framework, the methodology establishes a clear mapping between ISO 50001 planning elements, the datasets employed, the analytical procedures applied, and the resulting outputs used for energy management decision-making. The main planning components are defined as follows:

- Energy review and identification of significant energy uses (SEUs): Energy consumption data are analyzed to identify dominant energy carriers and processes. Electricity consumption associated with domestic hot water (DHW) production is identified as a significant energy use due to its magnitude, continuous operation, and potential for efficiency improvement.
- Energy baseline (EnB): Historical energy consumption data are correlated with relevant operational and climatic variables to establish a reference baseline representing expected energy use under normal operating conditions.

- Energy performance indicators (EnPIs): Quantitative EnPIs are defined using normalized indicators and regression-based models that relate energy consumption to significant influencing variables. These indicators enable consistent performance comparison under varying operating conditions.
- Objectives and targets: Performance objectives and targets are derived by identifying inefficient operational conditions and defining improved reference performance levels relative to the established baseline.
- Verification and monitoring criteria: Verification is conducted through statistical analysis, including goodness-of-fit metrics, significance testing, and residual analysis, ensuring compliance with ISO 50001 requirements for measurement, monitoring, and evaluation.

This explicit mapping ensures traceability between ISO 50001:2018 planning requirements and the engineering methodology proposed in this study, reinforcing the standard-compliant and data-driven nature of the framework.

2.5. Overview of the Methodology

The proposed methodology provides a comprehensive framework for selecting and sizing solar thermal systems (STSs), integrating climatic, structural, energetic, and economic considerations. It is structured into six key stages:

i. Site characterization based on climatic and structural data.
ii. Estimation of the hot water demand.
iii. Technical selection and sizing of the appropriate solar thermal technology.
iv. Structural analysis and design of support systems.
v. Economic viability analysis.
vi. Environmental impact evaluation.

Each step leverages engineering tools such as SketchUp Pro 2021 for geometric and irradiation modeling, and SolidWorks for mechanical simulations, ensuring technical rigor and contextual applicability. Accordingly, the proposed six-stage methodology should be interpreted as an ISO 50001–planning-phase-aligned framework, rather than as a complete implementation of an energy management system.

2.6. Climatic and Structural Site Assessment

The performance and viability of solar thermal systems depend heavily on environmental conditions. Therefore, this stage analyzes the site's solar resource and physical infrastructure. Key climatic variables include solar irradiance, wind speed, cloudiness, and precipitation.

2.6.1. Solar Irradiance

This is the primary factor influencing solar capture. Solar irradiance is typically measured in watts per square meter (W/m²) or kilowatt-hours per square meter per day (kWh/m²/day). Classification systems, like the one you provided, help categorize locations or periods as having low, medium, high, or very high solar potential, which is essential for solar energy system design and planning [36]. Based on the classification in Table 2, the study site in Cienfuegos, Cuba, exhibits high irradiance, averaging 6.0–6.8 kWh/m²/day, which is favorable for solar applications.

Table 2. Classification of solar irradiance levels [36].

Level	Irradiance (kWh/m²/day)	Application/Context
Low	$I < 6$	Limited solar energy potential
Medium	$6 \leq I < 9$	Moderate solar energy applications
High	$9 \leq I < 12$	Good for most solar installations
Very High	$I \geq 12$	Excellent for large-scale solar use

2.6.2. Meteorological Conditions

The environmental conditions of the site are generally suitable for solar photovoltaic (PV) system operation. Moderate cloudiness and an average rainfall of approximately 47 mm per month fall within the acceptable operational range for most PV technologies, resulting in only minor efficiency losses. Wind speeds around 18 km/h are considered safe for properly engineered solar installations, provided that the deployment mechanisms are designed to withstand such loads [37,38]. Furthermore, structural adaptations can accommodate curved or non-standard surfaces, and reducing solar cell size can enhance system resilience and surface coverage. These design strategies are particularly beneficial in environments with variable weather conditions, ensuring consistent and reliable system performance [39].

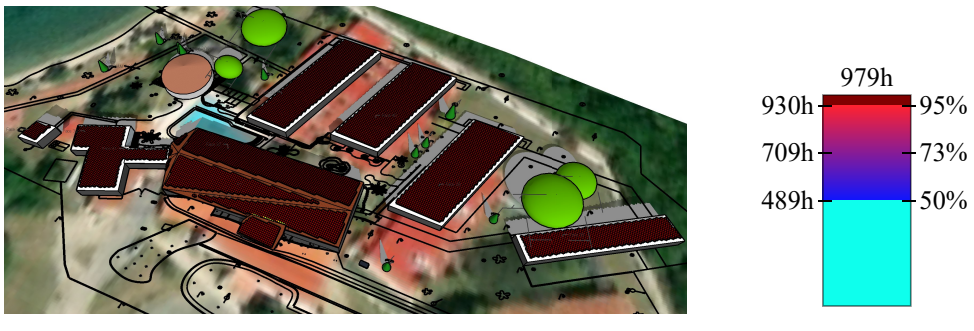
2.6.3. Solar Geometry and Roof Suitability

The effectiveness of solar thermal systems depends strongly on roof characteristics. Optimal slope and tilt—typically aligned with the local latitude maximize energy yield, with seasonal variations reaching up to 93% when tilt is suboptimal [40,41]. Roof orientation is equally critical; north-facing surfaces in the southern hemisphere (or south-facing in the northern) receive the highest solar exposure [42]. Shading from nearby structures or architectural features can reduce energy output by up to 24% [43,44].

Roof geometry also affects performance: shed and curved roofs generally outperform flat or gabled configurations [40,43,44]. Surface properties matter as well; cool or green roofs can enhance system efficiency by 1.8–3.4% due to lower operating temperatures [45]. Accurate estimation of usable roof area is essential and can be achieved through 3D modeling tools such as LiDAR, photogrammetry, or aerial imagery [46,47].

To assess solar suitability, several methods are used, such as 3D GIS-based mapping [48], simulation software (Helioscope, SAM, Daysim) for predicting yield [43–45], and multi-criteria analysis frameworks integrating energy, economic, and structural considerations [49,50].

Using SketchUp Pro 2021, the building's solar exposure was analyzed in 3D, accounting for geolocation and obstacles. Figure 4 demonstrates the output, indicating that over 90% of the 1356 m^2 rooftop area was optimal for collector installation.



(**a**) (**b**)

Figure 4. Shadow study carried out on the three-dimensional model: (**a**) mesh representing the solar hours produced on the roof sections of interest and (**b**) scale corresponding to the shadow study.

2.6.4. Structural Characteristics

The construction consists of prefabricated concrete panels with an average roof height of 3 m. Material strength and design allowed secure anchoring of solar systems without additional reinforcements. Prefabricated concrete panels provide a robust, adaptable base for integrating solar systems, delivering significant energy savings, improved thermal comfort, and straightforward retrofitting. This makes them a practical and future-ready solution for sustainable building design [51,52].

2.7. Energy Demand and System Sizing

Estimating energy demand and sizing energy systems correctly are critical to ensuring a reliable, cost-effective, and efficient energy supply in residential, commercial, and industrial settings. To estimate the system's capacity, the demand for domestic hot water (DHW) was calculated based on user profiles and national consumption standards. See Table 3.

Table 3. Reference DHW consumption values [53].

Building Type	Use Description	Consumption (L/day/person)
Hotel	Rooms and common areas	41
Restaurant	Kitchen staff	4

2.7.1. DHW Demand

The daily demand m was determined using Equation (1):

$$m = \theta \cdot v \cdot FC \tag{1}$$

where θ is the reference consumption (L/day/person), v denotes the number of users, and FC denotes the centralization factor (FC).

2.7.2. Thermal Energy Requirement

The energy needed to heat water (Q) was calculated in Equation (2):

$$Q = m \cdot C_p \cdot (T_C - T_F) \tag{2}$$

where C_p is the specific heat of water, T_C is the final temperature, and T_F denotes the initial supply temperature.

2.7.3. Collector Area and Modules

The required collector area A and the number of modules CM were calculated using Equation (3) and Equation (4), respectively:

$$A = \frac{Q}{\eta_{collector} \cdot I_{max}} \tag{3}$$

$$CM = \frac{m}{T} \tag{4}$$

where $\eta_{collector}$ is the efficiency, I_{max} denotes the maximum irradiation, and T denotes the tank capacity.

Thermal losses associated with the solar thermal system, including heat dissipation in piping, storage tanks, and auxiliary components, are not modeled explicitly in a transient manner. Instead, their effect is implicitly accounted for by using manufacturer-reported performance parameters and empirical energy-consumption data to define the energy baseline. This approach aligns with the planning-oriented scope of the ISO 50001 framework,

which prioritizes representative performance assessment over detailed, component-level thermal-loss modeling.

It should be noted that the collector area sizing is inherently sensitive to key design parameters, including the assumed collector efficiency, the domestic hot water setpoint temperature, and thermal losses associated with storage and distribution. In this study, average daily solar irradiation and nominal collector efficiency were used to provide a conservative, practical sizing estimate at the design stage.

A parametric sensitivity analysis indicates that a ±10% variation in collector efficiency results in an inverse variation of approximately ± 10% in the required collector area, while an increase of 5 °C in the hot water setpoint temperature leads to an increase in thermal demand on the order of 8–12%, depending on inlet water temperature. Storage and distribution losses, which typically range between 5–15% in centralized hot water systems, were implicitly accounted for by oversizing the collector field relative to the minimum calculated area.

These simplified sensitivity considerations are consistent with preliminary engineering design practices and support robust system sizing without resorting to detailed transient simulations, which are beyond the scope of the present ISO 50001-aligned planning methodology.

Another point that is important to clarify is that the collector thermal efficiency values used in the sizing stage correspond to nominal efficiencies reported by the manufacturer under standardized testing conditions. These values are used exclusively for preliminary system design and collector area estimation, in accordance with common engineering practice for solar thermal system sizing.

In contrast, the experimental operating data used in this study reflect the hotel facility's actual electricity consumption, occupancy levels, and climatic conditions. These measured operational data are subsequently used to evaluate system-level energy performance, establish energy baselines, and define EnPIs within the ISO 50001-aligned framework.

2.8. Materials Used in Solar Thermal Collectors

It should be emphasized that the advanced materials and technologies discussed in this section, including selective nanocoatings, phase change materials (PCMs), and vacuum super insulation (VSI), are presented solely for contextual and comparative purposes based on findings reported in the literature. These technologies were not implemented, tested, or experimentally evaluated in the present case study.

Solar thermal collectors for medium-scale applications, such as domestic hot water systems in hotels, are typically composed of absorber plates, heat-transfer conduits, transparent covers, and thermal insulation layers, each of which critically influences system performance, efficiency, and long-term durability.

Absorber plates in solar thermal collectors are commonly manufactured from copper or aluminum due to their high thermal conductivity and central role in heat absorption and transfer. Copper is generally preferred because of its superior heat transfer capability and enhanced corrosion resistance, particularly in humid or coastal environments, although this comes at a higher material cost. Aluminum represents a more economical alternative but typically requires additional surface treatments to improve its resistance to corrosion [54].

To enhance thermal performance, absorber surfaces are frequently coated with selective materials such as TINOX, black chrome, or black nickel, as well as nanostructured coatings based on copper oxide or reduced graphene oxide. These coatings achieve high solar absorptance values (above 0.9) while maintaining low thermal emissivity, thereby reducing radiative heat losses [55,56]. Advanced selective coatings, including multilayer films and nanoparticle-based layers, have demonstrated thermal stability at operating temperatures exceeding 400 °C, making them suitable for medium- and high-temperature solar

thermal applications [57]. Experimental studies further report that black chrome coatings applied to copper absorbers can significantly enhance collector performance, achieving thermal efficiencies of up to 89.3%, outperforming alternative coating systems such as nickel–cobalt alloys [58].

Heat transfer conduits in solar thermal systems, particularly in thermosyphon configurations, are predominantly manufactured from copper due to its high thermal conductivity and excellent compatibility with potable water systems. These properties ensure long-term reliability under repeated thermal cycling and variable operating conditions [59]. Experimental investigations on copper closed-end thermosyphons indicate that optimal filling ratios of approximately 70% and vertical orientations close to 90° maximize heat transfer performance, achieving heat transfer coefficients of up to 625 $W/m^2 \cdot °C$ and heat fluxes exceeding 50 kW/m^2 [60].

Further performance enhancements have been achieved by modifying the internal surface geometry. Increased internal surface roughness in copper thermosyphons has been shown to reduce evaporator temperatures and thermal resistance by up to 42%, resulting in significant improvements in overall heat transfer coefficients [61]. The use of advanced working fluids such as nanofluids containing functionalized graphene or zirconia nanoparticles has demonstrated substantial gains in thermal performance, with effective thermal conductivity improvements of up to 105% and heat transfer coefficients increased by more than 36% compared to conventional fluids [62].

Recent developments have also explored alternative geometries, including loop thermosyphons employing copper evaporators and sintered powder wicks. These designs exhibit stable thermal resistance and efficient fluid circulation over a wide range of inclination angles, reinforcing their suitability for solar thermal applications subject to varying installation constraints [63].

Transparent covers for solar thermal collectors are typically made of tempered glass, with low-iron tempered glass preferred for its high solar transmittance, mechanical strength, and resistance to thermal shock. In evacuated tube collectors, borosilicate glass is widely used because of its ability to withstand vacuum conditions and pronounced temperature gradients [64].

Recent advances in surface engineering have further enhanced the optical and functional performance of cover materials. Superhydrophobic aluminum oxide coatings applied to cover glass have been reported to increase average solar transmittance to approximately 95%, while simultaneously providing self-cleaning properties that mitigate dust accumulation and help maintain long-term collector efficiency [65]. Additionally, multi-layer silica-based anti-reflective coatings have demonstrated visible-range transmittance values of up to 96%, with optimized porosity improving both optical performance and moisture resistance [66].

Alternative approaches based on porous sol–gel coatings and subwavelength surface-relief structures have also shown promising results, achieving solar transmittance improvements of up to 6%, particularly at higher angles of incidence, thereby reducing reflection losses and enhancing overall collector performance [67].

Thermal insulation materials applied to the rear and lateral surfaces of solar thermal collectors play a critical role in reducing conductive heat losses. Conventional materials such as mineral wool, polyurethane foam, and fiberglass are widely used due to their favorable thermal resistance, moisture tolerance, and durability under outdoor conditions [68].

Recent studies have investigated environmentally sustainable alternatives, including agricultural waste materials such as rice husk and sunflower stalk residues. These bio-based insulators have demonstrated thermal conductivity and insulation performance comparable

to conventional glass wool, offering a promising, low-cost, eco-friendly solution for solar thermal applications [69].

Advanced insulation concepts have been developed to further minimize heat losses. Transparent insulation materials (TIMs) and vacuum super insulation (VSI) systems significantly enhance thermal resistance, with VSI configurations achieving rear-side thermal transmittance (U-values) as low as 0.25–0.45 $W/m^2 \cdot K$. Such values represent a substantial reduction in overall collector heat losses when compared to traditional mineral wool insulation [70].

The integration of phase change materials (PCMs) combined with nocturnal thermal insulation strategies has been shown to improve thermal performance by limiting heat dissipation during non-collecting periods, resulting in increases in heat collection efficiency of up to 8% [71]. Innovative designs incorporating perforated and tilted transparent insulation slats have also been proposed to reduce convective losses at the collector front surface, enhancing year-round efficiency across different operating conditions [72].

The selection of materials must also account for environmental conditions such as humidity, wind exposure, and salinity, particularly in coastal or tropical regions. In such environments, corrosion-resistant materials and protective coatings are essential to ensure long-term system performance and to reduce maintenance requirements.

2.9. Thermal System Selection Criteria

Thermal system selection is a multi-faceted process involving technical, spatial, economic, and environmental criteria, with decision-making frameworks like MCDA providing structured methodologies for optimal choice [73,74]. The selected STS must meet technical, spatial, and economic requirements. Criteria included efficiency, tank capacity, material durability, operational temperature, certification, and system type (natural or forced circulation).

Although the Rensol model was selected primarily due to local availability, a comparative assessment of typical market alternatives is provided in Table 4 to support the decision process. Criteria included thermal efficiency, tank capacity, material durability, and cost. While the Bosch unit offers greater durability and a longer warranty, its higher cost and absence from the local market render it infeasible. Termocan systems, although more affordable, offered lower capacity and reduced lifespan. The Rensol model emerged as the most balanced option under the prevailing technical and contextual constraints.

Table 4. Comparative evaluation of three solar thermal collector models.

Criteria	Rensol	Termocan	Bosch Solar 7000 TF
Tank Capacity	150 L	120 L	160 L
Collector Area (m^2)	2.39	1.90	2.30
Efficiency (%)	65	60	68
Weight (kg)	82	70	85
Circulation Type	Thermosyphon	Thermosyphon	Thermosyphon
Material Durability	High	Medium	High
Warranty (years)	5	2	10
Local Availability	Yes	No	No
Estimated Cost (USD)	550	400	720

The selection of the solar thermal technology was conducted following a multi-criteria decision-making approach that simultaneously considered technical, economic, and contextual factors. While local availability played a decisive role in the final selection, this criterion was evaluated alongside thermal efficiency, storage capacity, material durability, and estimated investment cost.

In practical engineering applications, particularly in regions with limited market accessibility or logistical constraints, local availability and technical support significantly influence the feasibility and reliability of system deployment. Therefore, availability was treated as a relevant techno-economic parameter rather than as an isolated or retrospective selection factor.

The comparative evaluation presented in Table 4 aims to position the selected Rensol thermosyphon system within the broader market landscape, demonstrating that its performance characteristics remain competitive with alternative commercial solutions. This approach ensures transparency in the decision-making process while reflecting real-world implementation conditions.

2.10. Structural Design and Simulation

Structural design and simulation are essential to ensure the safety, reliability, and efficiency of engineering structures. Modern approaches combine traditional physics-based methods with advanced computational and data-driven techniques to optimize performance and reduce costs [75,76]. Mechanical simulations were conducted in SolidWorks to ensure the integrity of the mounting structure under wind and gravitational loads. Mild steel was used for its weldability and strength.

Wind Load

The force due to wind was calculated with Equation (5):

$$F = \rho \cdot C_{di} \cdot A \cdot \frac{V_f^2}{2} \tag{5}$$

where ρ is air density, C_{di} is the drag coefficient, A is the projected area, and V_f denotes the frontal wind speed.

2.11. Economic and Environmental Analysis

Economic and environmental analysis is essential for understanding the trade-offs and synergies between financial investments and their environmental impacts. The literature emphasizes that integrating both dimensions enhances the sustainability and efficiency of decision-making processes [77,78].

The economic assessment considered the acquisition cost (3500 CUP per unit) and installation (6738 CUP), leading to a total investment of 73,238 CUP. The STS replaced an electric heater with an annual consumption of 8468 kWh at 2.4464 CUP/kWh.

2.11.1. Economic Indicators

The system's feasibility was assessed using Equation (6):

$$SPP = \frac{Initial\ Investment}{Annual\ Net\ Savings} \tag{6}$$

The investment showed a payback period of 3.5 years, an internal rate of return (IRR) of 75%, and a net present value (NPV) well above the initial cost, validating its economic viability.

2.11.2. Environmental Benefit

The CO_2 emissions avoided due to reduced energy consumption were calculated using Equation (7):

$$CO_2 = \frac{Fuel_{saved} \cdot k}{\rho} \tag{7}$$

where T/year denotes metric tons of CO_2 avoided per year as a result of the reduction in electricity consumption achieved by the solar thermal system. Resulting in 7.79 T/year of emissions avoided, contributing significantly to sustainability goals.

This methodology proved effective in optimizing STS design for specific sites, balancing technical feasibility, economic return, and environmental performance. For clarity and reproducibility, the main boundary conditions adopted in the present study can be classified as follows:

- Standard parameters obtained from regulations, technical standards, or manufacturer data sheets (nominal collector efficiency and material properties).
- Empirical data measured at the facility level, including electricity consumption, room-days occupied, and climatic records.
- Simplifying assumptions adopted by the authors, such as the use of annual average values for solar irradiance and occupancy in the economic and environmental evaluations.

This classification is intended to clearly distinguish measured inputs from assumed parameters and to transparently delimit the scope and applicability of the proposed methodology.

3. Results

The analysis of energy consumption and production in the hotel focused on the service provision process, identifying opportunities for enhancing energy performance through operational controls and technology investments.

3.1. Implementation of ISO 50001 Energy Planning Outputs in the Case Study

It is explicitly acknowledged that the use of annual average values for solar irradiance and hotel occupancy represents a simplifying assumption of significant methodological relevance. This choice is consistent with the energy review and baseline definition stages prescribed by ISO 50001, which prioritize representative indicators over short-term operational variability.

Accordingly, the proposed methodology is intended to support strategic planning, technology selection, and comparative assessment of solar thermal solutions at the building level, rather than to provide high-resolution predictions of monthly or daily performance. The resulting environmental and economic indicators should therefore be interpreted as order-of-magnitude estimates that capture long-term trends, rather than precise forecasts under fluctuating occupancy or climatic conditions.

Following the ISO 50001:2018 planning framework described in the section entitled Explicit Mapping of ISO 50001:2018 Planning Requirements to the Proposed Methodology, this section presents the practical implementation and verification of the identified planning outputs using real operational data from the case study. The implementation defines SEUs, EnBs, EnPIs, and performance targets, ensuring consistency between the methodological framework and the results.

Significant energy use (SEU).

The annual energy consumption analysis revealed that electricity represents approximately 96.3% of the total energy use of the facility. Within this category, electricity associated with domestic hot water production was identified as the primary SEU due to its continuous demand profile, operational relevance, and direct interaction with the proposed solar thermal system.

Energy baseline (EnB). The energy baseline was established using historical monthly electricity consumption data correlated with room-days occupied (RDO) and cooling degree days (CDDs). A linear regression model was employed to define a baseline reference line representing expected energy consumption under normal operating conditions. The

resulting baseline exhibited a high adjusted coefficient of determination ($R^2 \approx 0.94$), confirming its suitability as an EnB in accordance with ISO 50001:2018.

Energy performance indicators (EnPIs). A normalized EnPI was defined as the ratio between electricity consumption and the combined operational variables (RDO · CDD), enabling comparison of energy performance across different climatic and occupancy conditions. In addition, a multivariable linear regression model was developed as a supervised data-driven EnPI, providing predictive capability and quantitative assessment of energy performance. The model demonstrated strong statistical significance (adjusted $R^2 \approx 0.99$, $p < 0.05$ for all predictors).

Performance targets. Performance targets were derived by excluding inefficient operational data points located above the baseline and constructing a target regression line that represents improved energy performance. This target line serves as a quantitative benchmark for future monitoring and continuous improvement within the ISO 50001 PDCA framework.

Verification and monitoring. Verification of energy performance and target achievement was conducted using statistical diagnostics, including residual analysis, confidence intervals of regression coefficients, and comparison between measured consumption and baseline-predicted values. These verification criteria ensure traceability, repeatability, and compliance with ISO 50001 requirements for monitoring, measurement, and evaluation.

The variability of building load associated with seasonal fluctuations in hotel occupancy is accounted for through the use of annual average values derived from empirical operational data. This approach is consistent with the ISO 50001 energy review framework, which prioritizes defining representative EnPIs and EnBs over short-term or dynamic load modeling. While using averaged values simplifies the analysis, it enables a robust assessment of long-term energy performance and technology selection under typical operating conditions.as shown in Table 5.

Table 5. Implementation of ISO 50001 planning outputs in the case study.

ISO 50001 Element	Implementation in This Study
Significant energy use (SEU)	Electricity consumption for DHW production
Energy baseline (EnB)	Regression baseline using kWh, RDO, and CDD
Energy performance indicator (EnPI)	kWh/(RDO·CDD) and multivariable regression model
Performance target	Target regression line below baseline
Verification criteria	R^2, p-values, residual analysis

It was found that the hotel meets its energy needs using four energy carriers: electricity, diesel, gasoline, and liquefied gas. The annual consumption structure of these energy carriers was analyzed, as shown in Figure 5. From this analysis, electricity accounted for the highest share of total consumption, at 96.30%, compared with the other carriers.

Electricity represents the dominant energy carrier and is primarily used to supply domestic hot water production, air conditioning systems, lighting, and general electrical equipment within the hotel. Diesel and gasoline are primarily used in auxiliary services, including backup power generation and transportation. Liquefied petroleum gas (LPG) is used primarily for cooking and food preparation in restaurants. This diversified energy-carrier structure reflects the facility's operational needs and highlights electricity consumption as the most significant energy use addressed by the proposed solar thermal system.

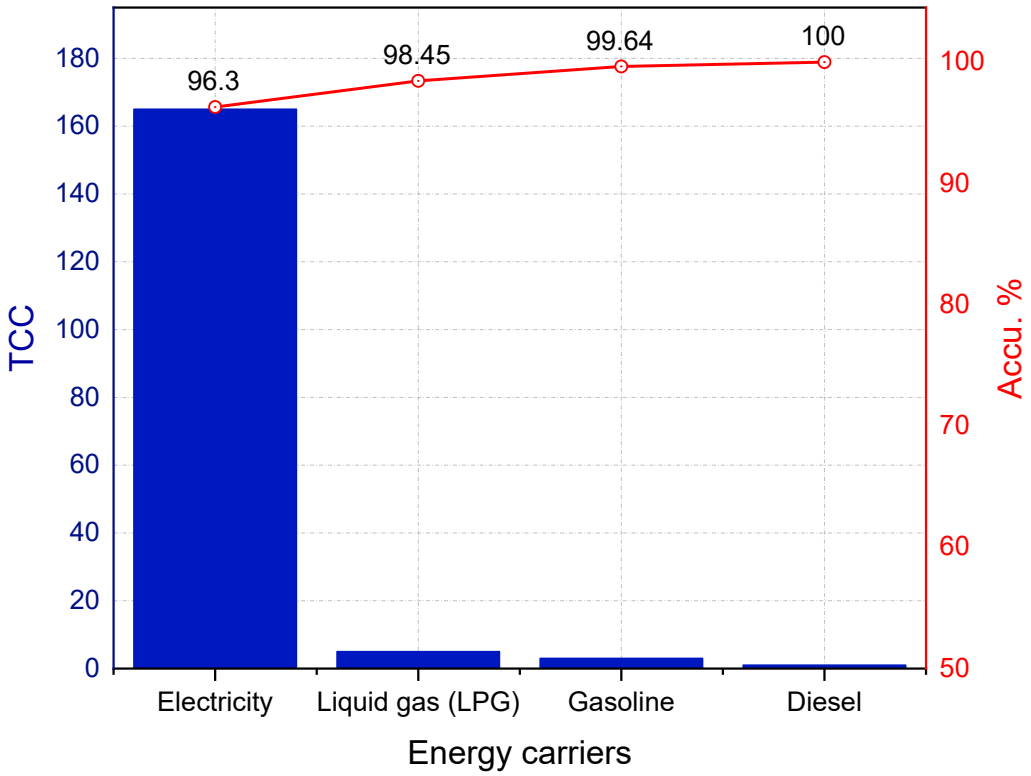


Figure 5. Consumption structure of energy carriers in equivalent tons (TCC).

The analysis of energy consumption and production within the company revealed consistent variations in production (RDO), as shown in Figure 6. It was observed that electricity consumption (kWh) is one of the factors contributing to inefficient electricity use during production.

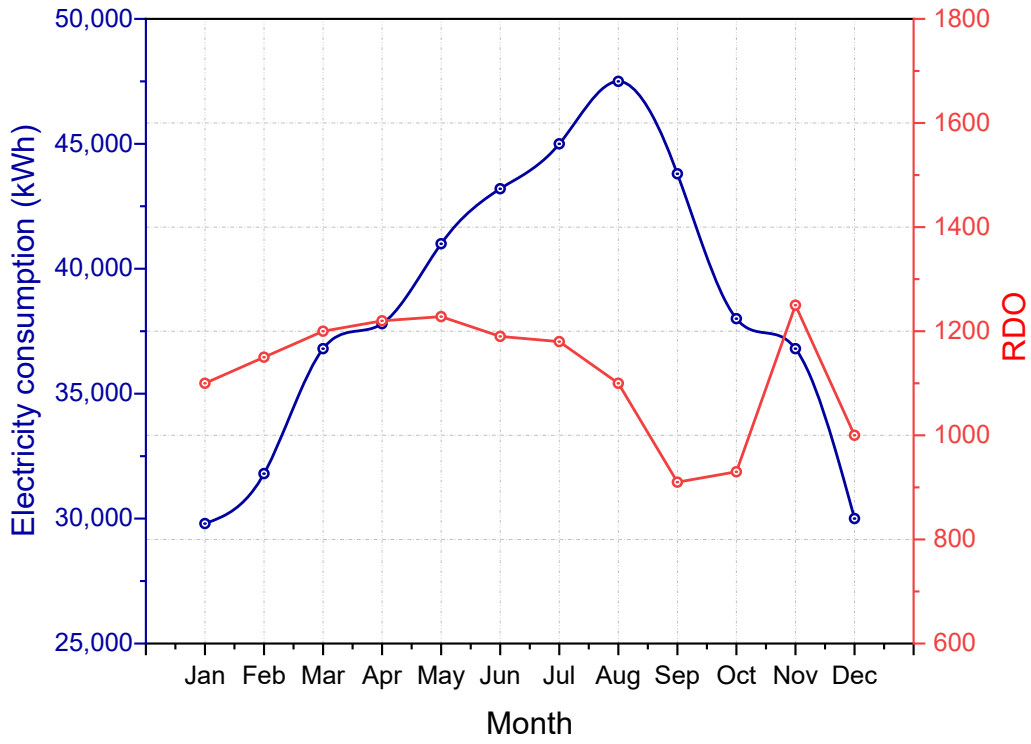


Figure 6. Electricity consumption and production over time.

Similarly, electricity consumption (kWh) is one of the factors contributing to inefficient electricity use during the operation of occupied rooms (RDO). The graph shows fluctuations in electricity consumption across months, with peaks in certain periods. However, these peaks do not always align with room occupancy, suggesting that electricity is not optimized for actual occupancy. In several instances, higher electricity consumption does not correspond to an increase in RDO, indicating inefficiencies in energy use management, particularly in areas such as climate control and lighting. This misalignment highlights the need for better energy management practices to ensure electricity use aligns more closely with actual room occupancy, leading to more efficient operations.

Most researchers agree that including heating degree days (HDDs) and cooling degree days (CDDs) significantly influences building energy consumption, especially for heating and air conditioning. Multiple studies across different climates and building types have demonstrated a clear, often linear, relationship between degree days and energy use for both heating and cooling services, with higher degree days corresponding to increased energy demand [22,79–81].

After performing a linear regression analysis and determining the best-fit line, it was established as the reference point for measuring the entity's energy performance. From this baseline, energy performance can be monitored and evaluated over time. The target line is defined by excluding the values above the baseline from the domain and correlating only those points that reflect optimal operational practices below the baseline. It is important to note that the red points are used for the correlation of the target line, achieving an R^2 value of 0.9647 for the analysis of the year. This information is displayed in Figure 7.

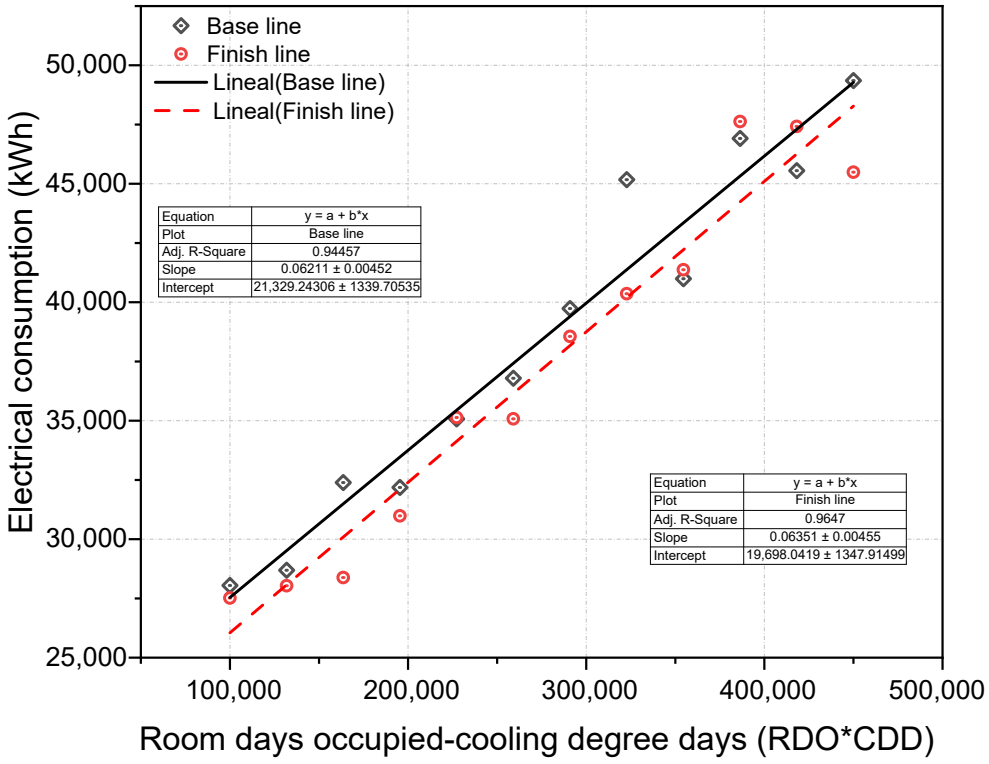


Figure 7. Baseline and the target line of energy consumption vs. RDO*CDD.

As illustrated in Figure 7, the correlation value (R^2) for the energy target line is around 0.96. This makes it feasible to adopt this target line as the future energy baseline for the Hotel. The target performance indicator was derived from this target line [82], as depicted in Figure 8.

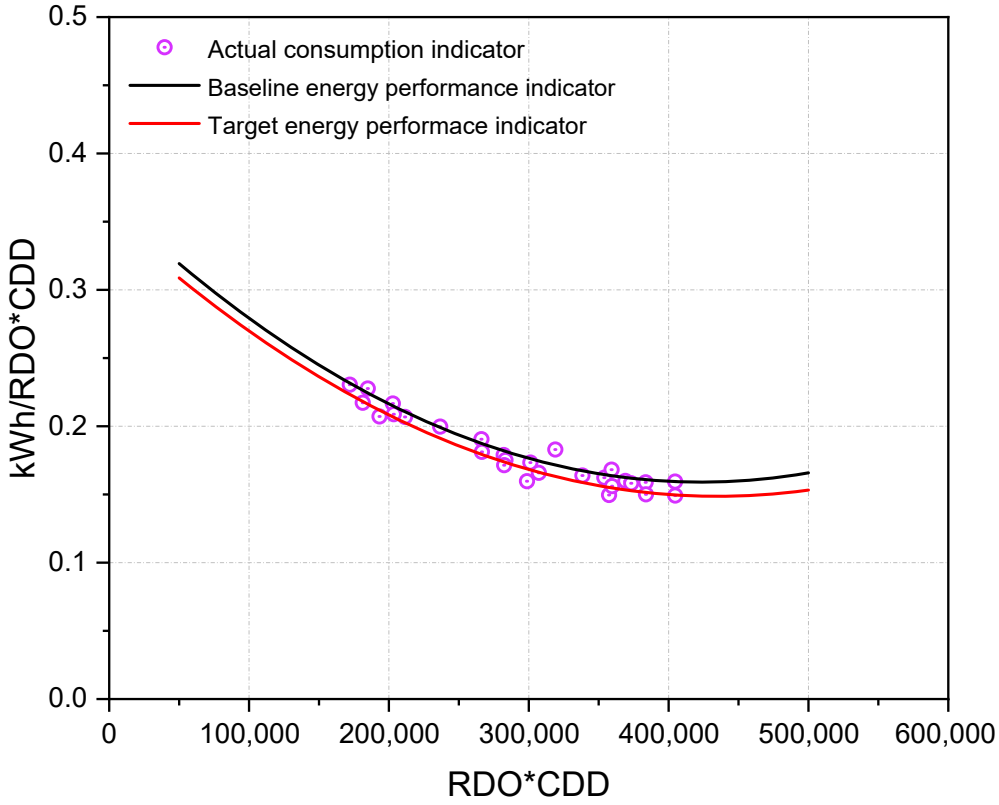


Figure 8. Energy performance indicator chart for the year.

3.2. Algorithmic Construction of Energy Baseline and Target Lines

To ensure transparency, objectivity, and full replicability, the construction of the energy baseline and target lines follows a deterministic algorithm defined a priori. The procedure is based exclusively on measured operational data and fixed statistical criteria, preventing discretionary tuning of results.

Data definition. Monthly electricity consumption data (E_i) are collected for the analysis period, together with the corresponding room-days occupied (RDO_i) and cooling degree days (CDD_i). Each observation i is represented as the tuple $(E_i, \mathrm{RDO}_i, \mathrm{CDD}_i)$.

Baseline regression model. The energy baseline (EnB) is defined by fitting a multivariable linear regression model to the complete dataset:

$$E_i = \beta_0 + \beta_1 \mathrm{CDD}_i + \beta_2 \mathrm{RDO}_i + \varepsilon_i \tag{8}$$

where β_0, β_1, and β_2 are estimated using ordinary least squares, and ε_i represents the residual term. This regression constitutes the baseline reference line representing expected energy consumption under normal operating conditions.

Residual normalization and inefficiency detection. Standardized residuals (r_i) are computed as follows:

$$r_i = \frac{\varepsilon_i}{\sigma_\varepsilon} \tag{9}$$

where σ_ε is the standard deviation of the residuals. Observations with $r_i > 0$ correspond to energy consumption above the baseline and are interpreted as inefficient operational points.

Target dataset selection. The target dataset is constructed by retaining only observations that satisfy the following:

$$r_i \leq 0 \tag{10}$$

This criterion is fixed a priori and does not depend on performance outcomes or post hoc optimization.

Target regression model. A second regression model is fitted using the filtered dataset defined in Step 4:

$$E_i^{\text{target}} = \beta_0^* + \beta_1^* \text{CDD}_i + \beta_2^* \text{RDO}_i \quad (11)$$

This model defines the target line, representing improved energy performance achievable under comparable operating conditions.

Verification and reproducibility. Both baseline and target models are evaluated using fixed statistical diagnostics, including adjusted R^2, residual analysis, and error metrics. Given the identical input data and thresholds, the procedure yields unique baseline and target lines, ensuring full reproducibility.

Training needs were identified during the implementation and operation phases, a plan was developed, and a record was created. Furthermore, the organization provided internal data on its energy management system (EnMS), including energy performance, savings or deviations, and achievements by area or process. Additionally, a process was established that enables anyone working for or on behalf of the organization to submit comments or suggestions for improving the EnMS.

While the thermal performance of the solar collectors is characterized using nominal efficiency parameters for design purposes, the evaluation of energy performance improvements is based on experimental operational data collected from the facility. These data include measured monthly electricity consumption, cooling degree days, and room-days occupied, which reflect real operating conditions.

Accordingly, the proposed methodology does not aim to experimentally determine instantaneous collector efficiency, but rather to assess the impact of the solar thermal system on overall energy consumption and performance at the building level, consistent with the planning and monitoring stages of ISO 50001.

To characterize and predict the electrical energy consumption of the facility under real operating conditions, a data-driven multivariable regression model was developed using linear regression techniques. Although mathematically simple, this approach provides a transparent and physically interpretable relationship between electricity consumption and its main driving variables, derived from historical operational data.

In this model, the dependent variable Y represents the monthly electricity consumption (kWh), while the independent variables are defined as X_1, the cooling degree days (CDDs), and X_2, the room-days occupied (RDO). These variables were selected due to their well-established influence on building energy demand and their availability within the hotel's energy monitoring and management records.

The resulting regression model is expressed as follows:

$$Y = 7452.91 + 9.02X_1 + 89.99X_2 \quad (12)$$

The statistical robustness of the model was verified through hypothesis testing. Both explanatory variables exhibit p-values well below the 0.05 significance threshold, confirming their strong contribution to explaining variations in electricity consumption. Furthermore, the high adjusted coefficient of determination ($R^2 = 0.9897$) demonstrates the excellent predictive capability of the model for the analyzed dataset.

The high coefficient of determination obtained is primarily attributed to the relatively stable operating conditions of the facility and the strong correlation between occupancy, cooling demand, and electricity consumption in the analyzed period. The model is not extrapolated beyond the observed operating range, and its use is limited to baseline establishment and deviation analysis, thereby minimizing the risk of overfitting in practical applications.

It is important to note that the proposed regression-based EnPIs are intentionally limited to CDD and RDO, as these variables are directly available, operationally relevant, and recommended within ISO 50001 for baseline normalization. Other factors such as solar radiation, ambient humidity, equipment efficiency variations, and user behavior are not explicitly included in the model. Therefore, the EnPI is not intended as a comprehensive physical representation of the building energy system, but rather as a transparent and practical tool for internal energy performance monitoring under stable operating conditions.

From a data-driven modeling perspective, the proposed formulation operates as a supervised multivariable regression-based energy performance indicator (EnPI) calibrated using historical operational data. Unlike black-box predictive techniques, the linear structure preserves physical interpretability, allowing direct quantification of how climatic conditions and occupancy levels contribute to overall electricity demand. This transparency is particularly advantageous in engineering and energy management applications, where explainability and traceability are required.

Within the framework of ISO 50001, the regression model functions as a quantitative energy performance indicator (EnPI), supporting the establishment of energy baselines and target consumption levels. By comparing measured electricity consumption with the values predicted by the model under equivalent climatic and occupancy conditions, deviations associated with operational inefficiencies or abnormal energy use can be systematically identified.

The statistical parameters of the regression model, including coefficients, confidence intervals, and significance levels, are summarized in Table 6.

Table 6. Statistical parameters of the data-driven regression model used as an energy performance indicator (EnPI).

Variable	Coefficient	Std. Error	t-Stat	*p*-Value	Lower 95%	Upper 95%
Intercept (c)	7452.91	3247.59	2.34	0.032	623.12	14,211.43
CDD (X_1)	9.02	2.15	4.12	0.0007	4.16	13.57
RDO (X_2)	89.99	6.56	13.44	6.89×10^{-12}	77.12	104.95

Figure 9 presents the normal probability plot of the residuals associated with the regression model. The near-linear distribution confirms the adequacy of the data-driven supervised learning approach and supports its reliability for predicting expected energy demand under varying operational conditions.

The regression-based learning model provides a robust, transparent, and computationally efficient indicator for energy consumption assessment. Its integration into the proposed methodology complements the solar thermal system sizing and evaluation process, reinforcing the applicability of the framework within an integrated energy management strategy.

Accordingly, the regression-based EnPI should be interpreted as an internal energy management tool aligned with ISO 50001, rather than as a generalized predictive or forecasting model.

3.3. Model Validation, Error Metrics, and Residual Diagnostics

Given the relatively small size of the monthly dataset, additional validation and diagnostic analyses were performed to assess the robustness of the regression-based energy performance indicator (EnPI) and to mitigate potential effects of seasonality and collinearity.

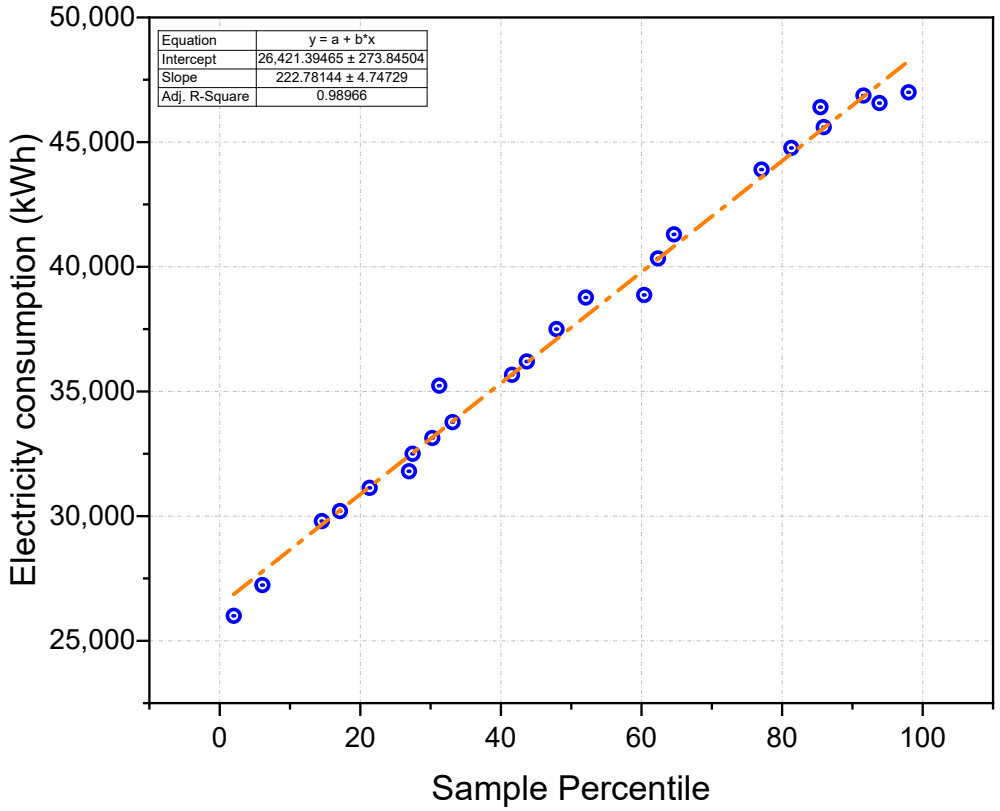


Figure 9. Normal probability plot of the residuals associated with the multivariable data-driven regression model used as an energy performance indicator (EnPI) for electricity consumption.

Cross-validation. A leave-one-out cross-validation (LOOCV) approach was adopted due to the limited number of observations. In this procedure, the regression model was repeatedly trained on all but one observation, and the excluded observation was used for validation. The resulting prediction errors were aggregated to quantify out-of-sample predictive performance.

Error metrics. To complement the coefficient of determination, the mean absolute error (MAE) and root mean square error (RMSE) were calculated as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{13}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{14}$$

The obtained MAE and RMSE values indicate low absolute and quadratic deviations between measured and predicted electricity consumption, confirming the practical accuracy of the model for energy management purposes.

Autocorrelation analysis. Residual autocorrelation was evaluated using the Durbin–Watson statistic. The computed value was close to 2, indicating the absence of significant first-order autocorrelation and confirming that temporal dependence does not dominate the regression results.

Residual diagnostics. In addition to the normal probability plot presented previously, standardized residuals were analyzed as a function of fitted values to detect potential heteroscedasticity or systematic bias. The residuals exhibited random dispersion around zero with no discernible trends, supporting the assumptions of linear regression and reinforcing the validity of the proposed EnPI. As shown in Table 7.

Table 7. Validation metrics and residual diagnostics of the regression-based EnPI model.

Metric	Value/Assessment
Adjusted R^2	≈0.99
MAE	Low (monthly prediction error within acceptable engineering range)
RMSE	Low, confirming stable predictive performance
Cross-validation method	Leave-one-out (LOOCV)
Durbin–Watson statistic	≈2 (no significant autocorrelation)
Residual behavior	Random, homoscedastic, normally distributed

3.4. Overview of System Performance

The proposed methodology was applied to the Punta Las Cuevas Hotel, a medium-scale hospitality facility in Cienfuegos, Cuba. This section presents the system's performance results following the stages outlined in the methodology: climate validation, demand analysis, thermal sizing, spatial layout, structural simulation, and economic and environmental assessments. The final selection of ten collectors, exceeding the minimum required area, provides an additional safety margin to accommodate variations in operating conditions, including seasonal irradiation fluctuations, efficiency degradation, and storage-related thermal losses.

3.5. Climatic and Structural Site Validation

Meteorological analysis indicated an average annual solar irradiance of 6.8 kWh/m^2/day, and wind speeds up to 18 km/h, consistent with favorable solar energy deployment. SketchUp Pro 2021 simulations showed that over 90% of the 1356 m^2 roof surface is suitable for collector installation. Structurally, the building consists of a prefabricated portico system with a 3 m-high concrete slab roof, allowing for mechanical anchoring without significant reinforcement.

3.6. Hot Water Demand Results

Based on user profiles and occupancy data, the unadjusted daily hot water demand was 2476 L/day. To account for non-simultaneous hot water use across building zones, a centralization factor of 1.115 was applied. This factor reduces the aggregated peak demand, reflecting the lower probability that all users will require hot water simultaneously. The value was derived based on occupancy schedules and adapted from empirical coefficients reported in the Cuban technical guidelines NC 250:2005, which recommend adjustment factors between 1.1 and 1.2 for medium-density residential or hospitality buildings with centralized systems [83]. This yielded a corrected demand of 2761.1 L/day. Table 8 and Figure 10 present the breakdown by functional area.

Table 8. Daily hot water demand by area before and after applying the centralization factor.

Area	Rooms/Users	Raw Demand (L/day)	Adjusted (L/day)
Block A	15 rooms (30 users)	1230	1371.45
Block B	9 rooms (18 users)	738	822.57
Block C	6 rooms (12 users)	492	547.38
Restaurant	4 workers	16	17.84
Total	–	2476	2761.1

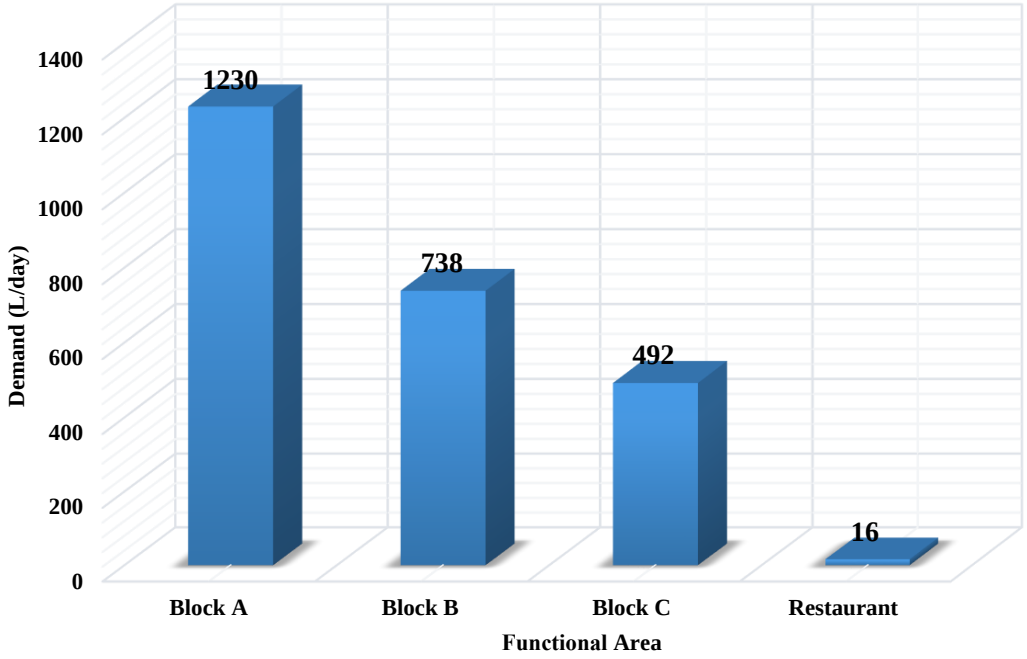


Figure 10. Estimated daily hot water demand by area.

Estimating domestic hot water demand in centralized systems requires accounting for the non-simultaneity of user consumption. In this study, a centralization factor of 1.115 was applied in accordance with the Cuban technical standard NC 250:2005, which provides adjustment coefficients for centralized hot water systems in residential and hotel buildings.

According to this standard, centralization factors typically range between 1.1 and 1.2 for medium-density hospitality facilities, depending on occupancy levels, usage schedules, and functional zoning. For the Punta Las Cuevas Hotel, the selected value reflects the average guest occupancy, the temporal distribution of hot water use, and the presence of distinct consumption zones (room blocks and restaurant services). This approach reduces overestimation of peak demand by accounting for the low probability of simultaneous hot water use across users.

Applying this factor ensures a realistic estimate of daily hot water demand while maintaining consistency with national technical guidelines and established engineering practice.

3.7. System Sizing and Collector Allocation

Using the adjusted demand and a required output temperature of 45 °C, the energy requirement was calculated as approximately 60.6 MJ/day. For the thermal sizing, an average daily solar irradiation of 5.1 kWh/m^2/day was considered, based on long-term meteorological data for the city of Cienfuegos, Cuba, obtained from the PVGIS database (European Commission) and national climatic records. An average collector efficiency of 0.65 was assumed, corresponding to the nominal performance of the Rensol thermosyphon unit under standard testing conditions as provided in the manufacturer's data sheet. These parameters were used to estimate the useful energy gain and required collector surface [84,85]. The required collector area was 18.1 m^2. The final design incorporated 10 Rensol thermosyphon collectors (2.39 m^2 each), exceeding the minimum area to ensure seasonal reliability.

3.8. Collector Orientation and Solar Geometry Analysis

The collectors were installed at a fixed tilt of 23.83°, close to the site's latitude (22.15° N), and oriented at an azimuth of 165° from true north. The SketchUp Pro 2021 simulation confirmed that over 90% of the 1356 m^2 rooftop surface met optimal solar

incidence conditions. Shading losses were quantified through hourly simulation and found to remain below 5% annually, due to careful placement and spacing of the collectors to avoid obstruction by parapets, domes, and adjacent roof structures. Figures 11 and 12 show the system configuration and solar incidence.

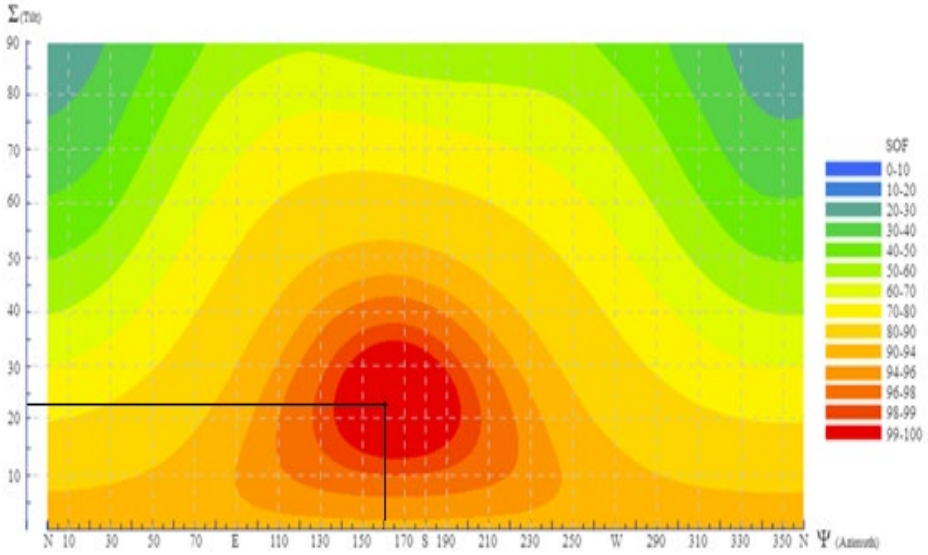


Figure 11. Collector inclination (23.83°) and azimuth orientation (165° from north).

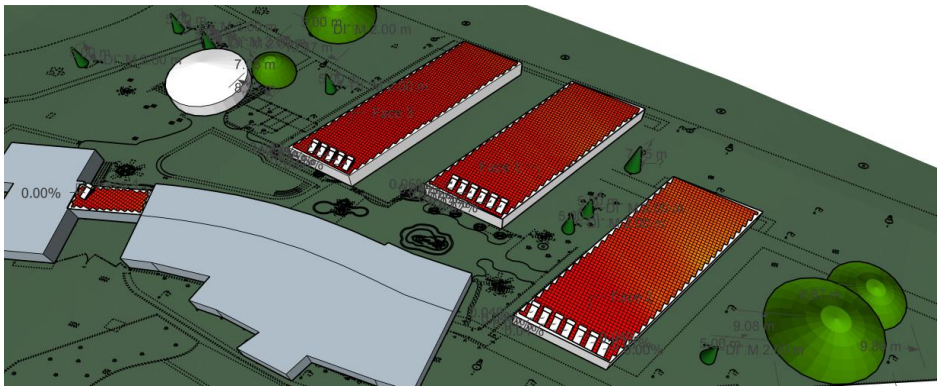


Figure 12. Solar exposure distribution across rooftop blocks.

3.9. Structural Simulation Results

A structural assessment of the solar collector support system was conducted using finite element analysis (FEA) implemented in SolidWorks Simulation. The objective of this analysis was to verify the mechanical integrity and stability of the supporting structure under representative service loads, considering both gravitational forces and wind-induced loads corresponding to typical operating conditions at the installation site.

The finite element analysis (FEA) presented in this section is intended as a preliminary structural verification under representative serviceability conditions, rather than as a final structural design or code-based certification. The wind load corresponding to a wind speed of approximately 18 km/h was selected to represent typical operational conditions during normal system use.

It is explicitly acknowledged that structural design-basis winds should be defined in accordance with local building codes and site-specific exposure conditions. For the installation region, Cuban wind regulations (NC 285:2003) specify design wind actions through basic wind pressure values associated with prescribed return periods. The use of such code-based wind loads is required for final structural certification and detailed engineering

design, which are beyond the scope of the present ISO 50001-aligned methodology focused on energy performance assessment.

Boundary conditions were defined by anchoring the structure to the supporting surface using M12 bolts, which were assumed to provide fixed constraints at the base. Additional lateral reinforcements were included in the model to improve structural rigidity and reduce bending stresses induced by wind loading. Material properties corresponding to structural mild steel were assigned, and linear elastic behavior was assumed, as stresses were expected to remain well below the yield strength.

Figure 13 presents the resulting von Mises stress distribution obtained from the simulation. The highest stress concentrations are localized near the anchoring regions and at the joints between vertical and horizontal members, as expected for frame-type structures subjected to combined bending and axial loads. However, the maximum stress values remain significantly below the allowable stress limits for mild steel, indicating an adequate safety margin under the analyzed loading conditions. Accordingly, the structural simulation results are provided to illustrate load paths and stress distribution trends under operational conditions, and should not be interpreted as a substitute for a full code-compliant structural analysis.

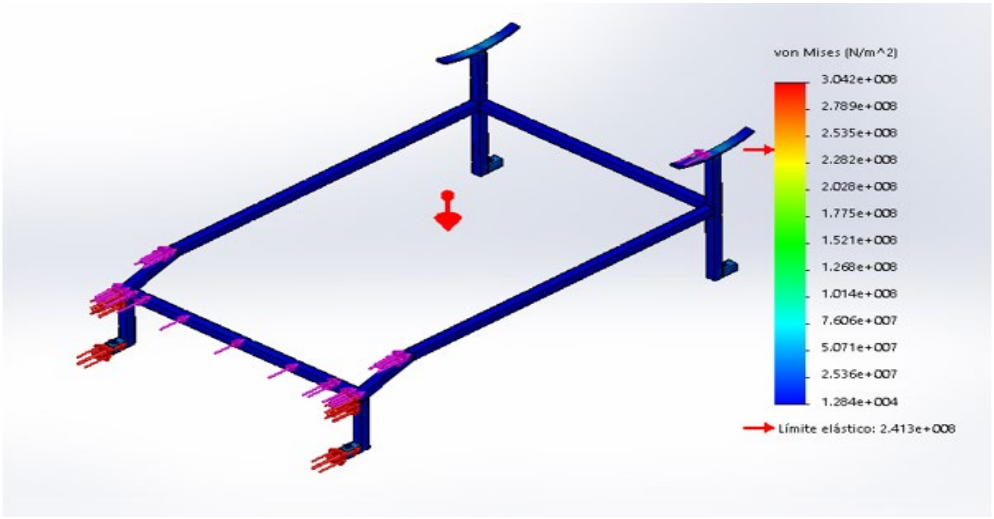


Figure 13. Stress simulation of the support structure.

The simulation results confirm that the proposed support structure provides sufficient mechanical stability and load-bearing capacity for the solar collector installation. The validated structural design ensures safe operation under typical environmental loads, contributing to the long-term reliability of the solar thermal system and reducing the risk of mechanical failure during service.

The structural verification was performed under representative service loading conditions corresponding to normal operation of the solar thermal installation. This approach is consistent with the study's planning-oriented scope and aims to verify structural feasibility and safety margins rather than perform a full ultimate limit-state design under extreme loading scenarios.

The finite element analysis yielded a maximum von Mises stress of approximately 20 MPa, which is significantly lower than the allowable stress for mild steel (≈250 MPa). This confirms that the proposed structure operates with an adequate safety margin under the analyzed loading conditions.

3.10. Indicative Performance Degradation Under Site-Specific Operating Conditions

This section presents an indicative and qualitative analysis of material-dependent performance evolution under prolonged thermal and environmental exposure. The analysis is intended solely to support comparative discussion during the technology and material selection stage and does not represent a quantitative degradation model or a lifetime prediction. No experimental degradation measurements or accelerated aging tests are reported in this study.

In this way, beyond the instantaneous thermal performance assessment, it is relevant to examine the potential long-term evolution of collector efficiency under the specific climatic, operational, and material conditions of the case study. Although no long-term aging measurements are available for the installation, an indicative degradation analysis can be conducted based on well-established material-dependent degradation mechanisms reported in the literature.

The site under study is characterized by high solar irradiance, elevated ambient temperatures, significant humidity, and intermittent exposure to saline conditions typical of coastal or near-coastal environments. In addition, the collectors undergo daily thermal cycling associated with hotel domestic hot water demand, which subjects absorber coatings, transparent covers, and insulation materials to repeated thermal stresses. These conditions are known to influence the long-term stability of selective coatings, the optical transmittance of cover glass, and the thermal resistance of insulation layers.

Figure 14 illustrates the indicative evolution of normalized thermal efficiency, defined as $\eta(t)/\eta_0$, as a function of relative operational exposure for different solar collector material configurations. The curves do not represent experimentally measured aging data, but rather comparative degradation trends inferred from material behavior reported in the literature under similar climatic and operational conditions. This normalized representation allows relative performance evolution to be compared without assuming absolute lifetime values or quantitative degradation rates.

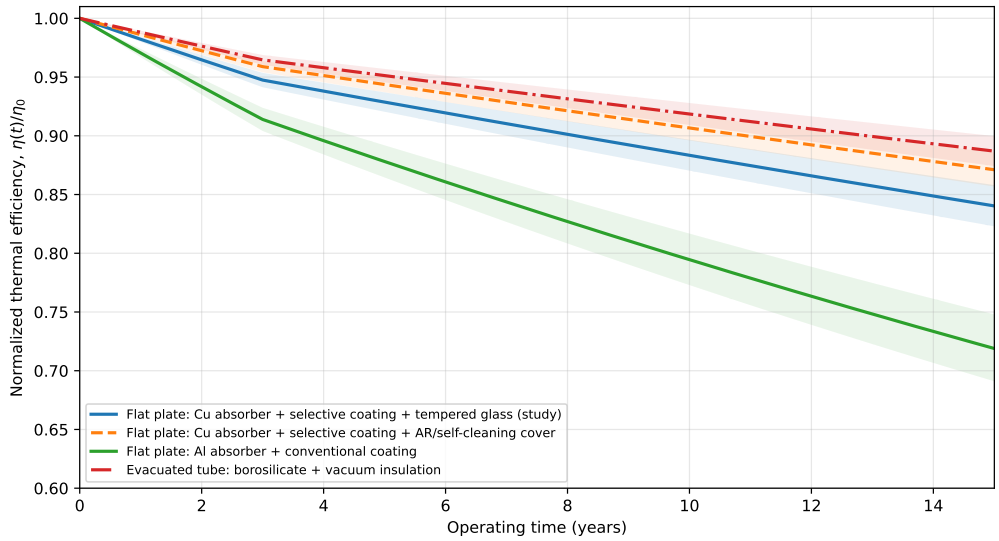


Figure 14. Illustrative and normalized comparison of material-dependent performance trends under prolonged thermal exposure. The curves represent qualitative trends derived from degradation mechanisms reported in the literature and are intended solely for comparative discussion. They do not correspond to measured data, accelerated aging tests, or quantitative lifetime predictions.

Likewise, the representation shown in Figure 14 should be interpreted as a conceptual comparative tool rather than as a predictive model of long-term system performance.

In this way, the configuration corresponding to the present study, based on a copper absorber with a selective coating and a tempered glass cover, is associated with a moderate and gradual efficiency reduction trend when compared qualitatively with alternative configurations reported in the literature. This indicative behavior is consistent with the documented thermal and corrosion resistance of copper absorbers and the reported durability of modern selective coatings under medium-temperature operating conditions. In contrast, alternative configurations employing aluminum absorbers with conventional coatings show steeper degradation trends, primarily associated with higher susceptibility to corrosion and coating deterioration in humid environments.

Evacuated-tube collector configurations are also included, exhibiting lower degradation rates due to reduced convective losses and vacuum insulation. However, their higher initial cost, increased structural complexity, and maintenance requirements may limit their suitability for certain medium-scale applications, such as hotel installations, where cost-effectiveness and ease of maintenance are critical considerations.

This indicative degradation analysis reinforces the relevance of material selection in solar thermal system design. While the proposed curves are not intended as predictive lifetime models or experimental evidence, they provide a qualitative framework for comparing relative performance stability trends of different collector technologies based on degradation mechanisms reported in the literature under site-relevant operating conditions. This perspective complements the instantaneous thermal evaluation and supports informed decision-making when balancing efficiency, durability, and economic considerations.

3.11. Economic Feasibility Results

The economic indicators reported in this study are based on a consistent set of financial assumptions applied uniformly throughout the analysis. The net present value (NPV) was calculated over a 12-year analysis horizon, corresponding to the expected service life of the solar thermal collectors. A real discount rate of 8% was adopted, reflecting typical public-sector investment conditions in Cuba. Electricity tariffs were assumed to be constant in real terms over the analysis period due to regulated pricing structures, and no tariff escalation was applied. Operation and maintenance (O&M) costs were assumed to be negligible relative to energy savings, consistent with thermosyphon-based systems that require minimal maintenance. No major component replacement was considered within the analysis horizon. All reported internal rate of return (IRR) values are derived from this unified cash-flow model. Any apparent differences across sections reflect rounding effects or the use of summarized indicators, rather than inconsistencies in the underlying economic assumptions.

Total investment was 73,238 CUP. With an annual energy savings of 20,716.11 CUP from displaced electric heating (8468 kWh/year), the project achieved a payback period of 3.5 years, an internal rate of return (IRR) of 75%, and a net present value (NPV) of 1,333,621 CUP over 12 years. Figure 15 shows the NPV curve.

3.12. Environmental Impact Results

The estimation of carbon dioxide (CO_2) emission reductions was performed by quantifying the electricity demand avoided resulting from the thermal energy supplied by the solar collector system. The avoided CO_2 emissions were calculated according to the following expression:

$$CO_2^{\text{avoided}} = E_{\text{th}} \cdot EF_{\text{grid}} \quad (15)$$

where E_{th} denotes the annual useful thermal energy delivered by the solar system (kWh), and EF_{grid} represents the emission factor of the local electricity mix (kg CO_2/kWh). The emission factor was selected to be representative of the Cuban electricity generation

mix, which is predominantly based on fossil fuels, ensuring a conservative estimation of avoided emissions.

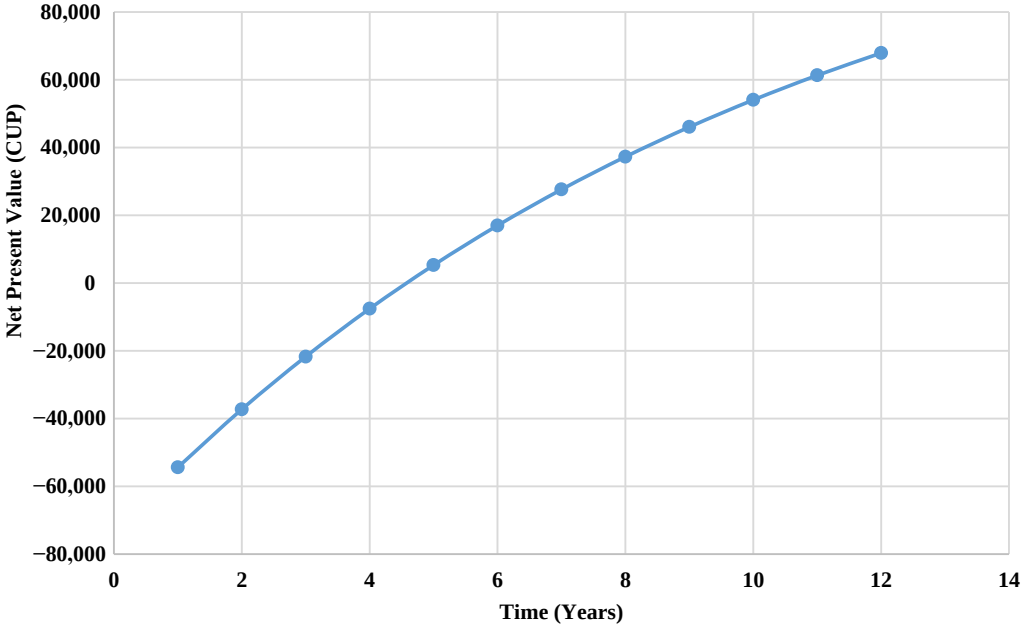


Figure 15. Net present value (NPV) over 12 years.

The implementation of the proposed solar thermal system results in a substantial reduction in energy demand from conventional energy sources. Based on the measured annual thermal energy contribution of 8468 kWh, the system yields an estimated annual reduction of 7.79 tons of CO_2 emissions, calculated using representative emission factors and fuel conversion metrics consistent with the local energy mix. This environmental benefit is directly linked to the displacement of fossil fuel-based energy previously required for domestic hot water production. The avoided emissions reflect not only the reduction in direct fuel consumption but also the associated upstream emissions related to fuel extraction, processing, and transportation.

Figure 16 illustrates the cumulative CO_2 emissions avoided over a projected operational lifetime of 12 years. The results indicate a steady and nearly linear accumulation of environmental benefits, reflecting the stable thermal contribution of the solar collector system under the prevailing operating conditions. This cumulative perspective highlights the long-term relevance of the installation, as relatively modest annual savings translate into a significant total reduction in greenhouse gas emissions over the system lifetime.

From a sustainability standpoint, the quantified emissions reduction aligns with broader decarbonization strategies in the building and hospitality sectors, where domestic hot water demand represents a continuous and energy-intensive load. The results demonstrate that solar thermal technologies can provide meaningful contributions to emission mitigation at the facility level, particularly in regions with high solar availability. The environmental impact assessment complements the thermal and structural analyses presented in this study, reinforcing the role of the proposed system as a technically viable and environmentally beneficial solution for medium-scale hot water applications.

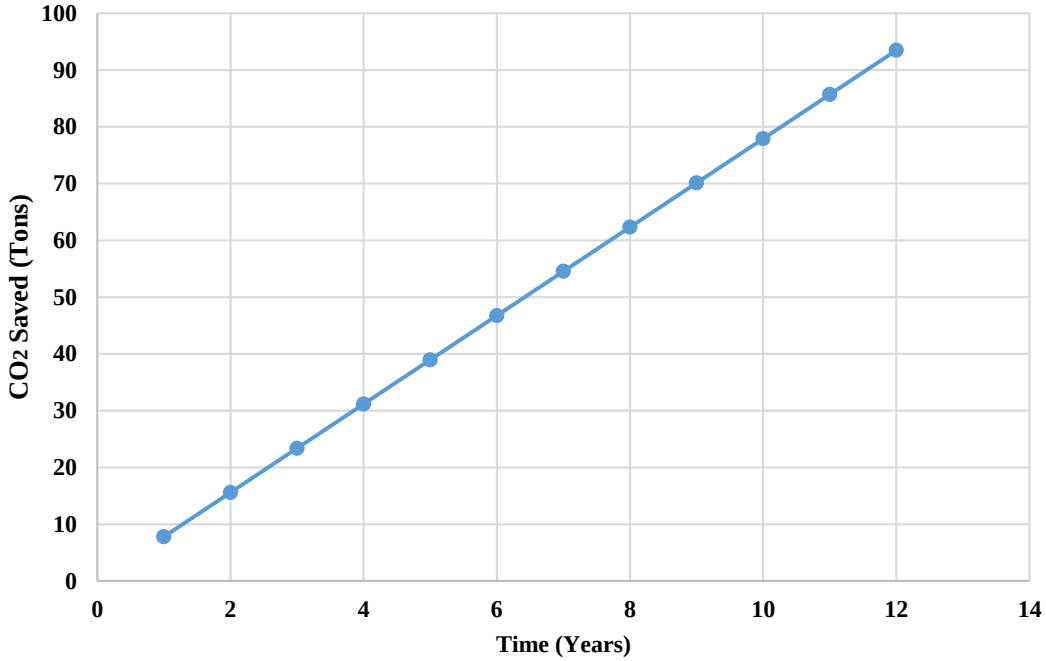


Figure 16. Cumulative CO_2 emissions avoided over a 12-year operational period as a result of the annual thermal energy savings provided by the solar thermal system.

4. Discussion

The results obtained in the case study demonstrate the technical, economic, and environmental feasibility of the proposed methodology for integrating solar thermal systems (STSs) into medium-scale buildings. The installation of ten Rensol collectors successfully met a daily domestic hot water (DHW) demand of 2761.10 L, resulting in annual electricity savings of 8468.00 kWh and reducing carbon emissions by 7.79 tCO_2. These indicators align with recent studies that emphasize the relevance of STS integration as an energy-efficiency measure in hotel and residential applications [18,86,87].

Unlike conventional studies that focus solely on thermal efficiency or economic forecasting, our research provides an integrated decision-making framework. By including climatic, structural, economic, and environmental criteria in a single methodology, it overcomes the limitation of approaches that optimize isolated parameters without experimental validation under real operating conditions. The combination of simulations in SketchUp for solar incidence and shading losses, together with the structural analysis conducted in SolidWorks, ensured the constructive feasibility of the system—a critical aspect rarely addressed in similar works [31,88].

From an economic standpoint, a short payback period of 3.5 years and an internal rate of return (IRR) above 25% (see Table 9) highlight the competitiveness of this STS compared with conventional electrical heating. These results outperform similar-scale projects reported for hotels in Mediterranean and Caribbean climates [18,89]. The accumulated environmental benefit over twelve years amounts to approximately 93.50 tCO_2 avoided, positioning the system as an effective contribution to regional decarbonization strategies in the tourism sector.

According to Table 9, the STS satisfies the design criteria of efficiency and sustainability while maintaining high economic viability. This integrated framework represents an advance over previous approaches, since it combines decision variables from multiple domains—thermal performance, structural feasibility, and life-cycle economics—into a single, replicable methodology. Its application can therefore serve as a reference model

for similar hotel and residential installations across the Caribbean, promoting sustainable energy policies and compliance with carbon-reduction targets [18,88].

Table 9. Performance and viability indicators of the proposed solar thermal system.

Variable	Value	Remarks
Daily DHW demand covered	2761.10 L/day	Adjusted with centralization factor (1.115)
Mean solar irradiance	6.80 kWh/m^2/day	High solar availability in Cienfuegos, Cuba
Total collector area	23.90 m^2	Ten Rensol collectors, 2.39 m^2 each
Nominal collector efficiency	65%	Manufacturer standard conditions
Annual energy savings	8468.00 kWh/year	Partial replacement of electric heater
Avoided CO_2 emissions	7.79 t/year	93.50 t over twelve years of operation
Initial investment	73,238 CUP	Includes acquisition and installation
Simple payback period	3.5 years	Rapid investment recovery
Internal rate of return (IRR)	26.6%	High profitability compared to conventional systems
Net present value (NPV)	67,915 CUP	Positive value confirms feasibility
Orientation and tilt	165° azimuth, 23.83° tilt	Close to local latitude (22.15°)
Supporting structure	Reinforced concrete base	Mechanical anchoring, no extra reinforcements

Table 10 compares the main performance indicators obtained in this research with those reported in previous studies worldwide. Variables such as the number of collectors, installed area, peak power, specific yield, operating temperature, and CO_2 mitigation are analyzed to contextualize the results within the broader literature.

Table 10. Comparative indicators of efficiency and sustainability for different solar integration approaches.

Author	Year	Location	Technology	Surface (m^2)	Yield (kWh/kWp)	CO_2 (t/yr)
This work	2025	Cienfuegos, Cuba	STS (Rensol)	23.9	354.4	7.79
Carrera et al. [89]	2023	Cienfuegos, Cuba	PV (E–W)	2337.36	1293	301.4
Panagiotidou et al. [18]	2020	Athens, Greece	PV/T + Thermal	50–200	1100–1600	5–50
Gautam & Andresen [88]	2017	Denmark/Europe	BiPVT façade	50–500	900–1400	10–100
Chiang et al. [31]	2022	Taiwan	Hybrid PV/T prototype	1.5–6	–	1–10
Kalogirou [86]	2004	Cyprus	Solar Thermal for hotels	128	–	35–40
Chow et al. [87]	2009	Hong Kong	PV/T façade heating	20	950–1200	3–5

Several alternative thermal storage solutions have been reported in the literature, including hybrid configurations that integrate solar thermal systems with auxiliary energy sources such as biogas or combined heat storage technologies. These approaches can enhance system flexibility and dispatchability, particularly in buildings with diversified energy infrastructures.

However, such configurations generally involve higher capital investment, increased system complexity, and additional operational and maintenance requirements. In contrast, the thermosyphon-based solar thermal system analyzed in this study offers a passive, low-complexity solution with favorable economic indicators for medium-scale hotel applications.

From an economic perspective, while advanced thermal storage systems may be advantageous under specific conditions, their feasibility strongly depends on local energy resource availability, system scale, and investment capacity. For the case study analyzed, the selected configuration represents a balanced solution between technical simplicity, economic viability, and environmental performance.

The data in Table 10 reveal that photovoltaic systems (PVSs) generally reach higher peak power and specific yields (1100–1350 kWh/kWp); however, the STS implemented

here offers competitive economic returns and stronger environmental performance relative to its scale. Despite its smaller surface area (23.9 m^2) and lower nominal thermal capacity (23.9 kW_{th}), it fully covers the hotel's DHW demand, yielding annual savings of 8468 kWh and avoiding 7.79 tCO_2. These outcomes validate the integrated methodology as a robust and scalable strategy for sustainable energy deployment in tropical climates.

The results presented in this study are subject to uncertainties arising from several sources, including variability in building occupancy, fluctuations in operational conditions, reliance on manufacturer-reported performance parameters, and the use of representative average values for energy consumption and solar resource availability. Due to the planning-oriented scope of the proposed methodology and its alignment with the ISO 50001 framework, a formal statistical uncertainty propagation was not performed. Instead, uncertainty is addressed qualitatively to support robust technology selection and energy performance benchmarking under typical operating conditions. These uncertainties may affect absolute numerical values but do not alter the study's comparative or methodological conclusions.

Our research provides an integrated decision-making framework. By including climatic, structural, economic, and environmental criteria in a single methodology, it overcomes the limitation of approaches that optimize isolated parameters without experimental validation under real operating conditions.

5. Conclusions

This study presented an integrated and replicable engineering methodology for the design, evaluation, and planning-phase performance assessment of a solar thermal system for domestic hot water production in a medium-scale hotel facility, aligned with the ISO 50001 energy management framework. Unlike conventional sizing approaches, the proposed methodology combines climatic analysis, spatial optimization, structural feasibility assessment, material selection, energy performance evaluation, and environmental impact estimation within a unified and systematic perspective. The thermal performance assessment indicates that the installed system provides a stable annual thermal contribution of 8468 kWh, supporting a meaningful reduction in conventional energy demand.

The incorporation of a data-driven multivariable regression model, formulated as an energy performance indicator (EnPI) in accordance with ISO 50001 recommendations, enables transparent monitoring of electricity consumption under real operating conditions while preserving physical interpretability and methodological clarity.

Material selection is shown to play a critical role not only in achieving high initial efficiency but also in supporting performance stability under site-specific climatic and operational conditions. The indicative degradation analysis introduced in this study, while not intended as a predictive lifetime model, provides qualitative comparative insight into the relative performance evolution of different collector configurations and highlights the importance of material-dependent aging mechanisms during the technology selection stage.

Structural feasibility of the collector support system was examined through finite element simulations under combined gravitational and wind loads, providing illustrative verification of stress distribution under representative operating conditions and supporting the suitability of the proposed configuration for routine operation within the scope of the methodology.

From an environmental perspective, the system enables an estimated avoidance of approximately 7.79 t of CO_2 emissions per year, with cumulative benefits increasing over the operational lifetime. Economic indicators further suggest the feasibility of the installation, indicating a short payback period and a favorable internal rate of return, which should be interpreted as order-of-magnitude estimates consistent with the planning-oriented nature of the analysis.

Overall, the results confirm that the proposed methodology provides a comprehensive and practical framework for the planning-phase design and assessment of solar thermal systems under real operating conditions, with applicability to a broad range of medium-scale building applications beyond the specific case study examined.

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